

Python for Data Science

Data Science / AI Fundamentals

Why Python for Data Work

Python has become the default language for data science and AI largely because of its ecosystem, not the language itself — libraries like NumPy, Pandas, and scikit-learn provide fast, well-tested implementations of the operations data work needs constantly, and its readable syntax makes it easy to express data transformations clearly. A data scientist rarely writes performance-critical low-level code directly in Python; instead, they use these libraries, which are themselves implemented in faster languages underneath (much of NumPy and Pandas' core is written in C), getting both Python's readability and near-native speed for the actual number-crunching.

Python Fundamentals for Data Work

The Python fundamentals that matter most for data work: lists and dictionaries as the basic data structures (before you ever touch a library-specific structure like a DataFrame), list comprehensions for concise data transformations, and functions for making repeated analysis steps reusable. Understanding Python's core iteration and data structures deeply pays off constantly, since even when working inside Pandas or NumPy, you're frequently converting between their specialized structures and plain Python lists/dicts at the boundaries of an analysis.

Working with Files & Data Formats

Real-world data rarely arrives as a clean, ready-to-use structure — it arrives as files, and the first job is reading them correctly:

- CSV — the most common tabular format; comma-separated values, usually with a header row naming each column.
- JSON — nested, hierarchical data, common for API responses and semi-structured records.
- Excel — common in business contexts, often with multiple sheets and merged/formatted cells that complicate clean parsing.
- Understanding encoding issues (files not being plain UTF-8), missing values represented inconsistently (empty string vs. 'NA' vs. 'null' vs. an actual blank), and malformed rows are practical realities of working with files that no tutorial dataset prepares you for.

Scripting for Automation

A large part of practical data science work is automation — running the same cleaning, transformation, or reporting steps repeatedly, reliably, on new data as it arrives, rather than manually repeating steps in a notebook every time. Writing analysis as reusable functions and scripts (rather than one long sequence of ad-hoc notebook cells) makes it possible to re-run an entire pipeline on new data with one command, schedule it to run automatically, and hand it off to someone else without them needing to understand every intermediate step by reading your mind.

Learn More

Python Official Documentation

<https://docs.python.org/3/>

Pandas Official Documentation

<https://pandas.pydata.org/docs/>

freeCodeCamp YouTube Channel

<https://www.youtube.com/@freecodecamp>